Task 21 — Cost Optimization & FinOps

Data Analyst · Sprint E - Hardening, Compliance & Go-Live · Task Brief

DATA ANALYST PlaceMux · Phase 3 · Task Brief

TASK 21 / 25 **Principal / Certification · Principal** Sprint E - Hardening, Compliance & Go-Live · Mastery 93%

Focus

Build the unit-economics model: cost to serve, contribution margin and whether scaling is profitable.

What to build / complete

- A unit-economics model (cost per application, shortlist, placement; CAC; LTV)
- Contribution margin by segment and tenant
- A profitability view of growth at scale

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: Leadership knows whether growing PlaceMux makes money or loses it faster - per segment.

Stage B — Build: A unit-economics model (cost per application, shortlist, placement; CAC; LTV)

1. Define the metric, its source & the decision

Define precisely what “A unit-economics model (cost per application, shortlist, placement; CAC; LTV)” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: Contribution margin by segment and tenant

1. Define the metric, its source & the decision

Define precisely what “Contribution margin by segment and tenant” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: A profitability view of growth at scale

1. Define the metric, its source & the decision

Define precisely what “A profitability view of growth at scale” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Present the unit economics with every cost input traced to real data.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “A unit-economics model (cost per application, shortlist, placement; CAC; LTV)” is complete, real, and demoable end-to-end.
- “Contribution margin by segment and tenant” is complete, real, and demoable end-to-end.
- “A profitability view of growth at scale” is complete, real, and demoable end-to-end.
- Good looks like: Leadership knows whether growing PlaceMux makes money or loses it faster - per segment.

Worry if any of these are true

- Unit economics ignoring infrastructure and AI cost.
- Blended averages hiding an unprofitable segment.
- Calling an experiment early because it looked good on day two.

Hand-off

- Unit economics to leadership and DevOps. Make sure the next person can pick it up without you.